Data and Artificial Intelligence

Cyber Shujaa Program

Week 10 Assignment
Deep Learning

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Contents

Data and Artificial Intelligence 1
Cyber Shujaa Program 1
Week 10 Assignment Deep Learning 1
Introduction 2
The key objectives of the assignment were to: 2
Tasks Completed 2
Task 1: Load and Explore the Dataset 2
Task 2: Data Preparation and Preprocessing 3
Task 3: ANN Model Construction 5
Task 4: Model Compilation 6
Task 5: Model Training 7
Task 6: Model Evaluation 8
Task 7: Training Visualization 9
Task 8: Confusion Matrix 10
Task 9: Classification Report 11
Task 10: Model Saving and Reloading 12
Conclusion 12

Introduction

The objective of this assignment was to develop practical experience in building a deep learning model using Artificial Neural Networks (ANN) with TensorFlow/Keras for image classification.

The MNIST dataset was used, which contains grayscale images of handwritten digits (0–9). Each image is 28×28 pixels, making it a standard benchmark dataset for evaluating classification models.

The assignment focused on key deep learning concepts such as data preprocessing, neural network architecture design, model training, evaluation, and performance visualization.

The goal of the project was to build an accurate classification model capable of recognizing handwritten digits while demonstrating best practices in deep learning workflows.

The key objectives of the assignment were to:

1. Load and explore the MNIST dataset
2. Preprocess image data for neural network training
3. Build an ANN model using Keras Sequential API
4. Train and validate the model
5. Evaluate model performance using test data
6. Visualize training results
7. Generate classification metrics and confusion matrix
8. Save and reload the trained model

Tasks Completed

Link to Code via Google Colab:

https://colab.research.google.com/drive/1p8LjURtMT_SElkrLBwmV0blfG4GG3JT9?usp=sharing

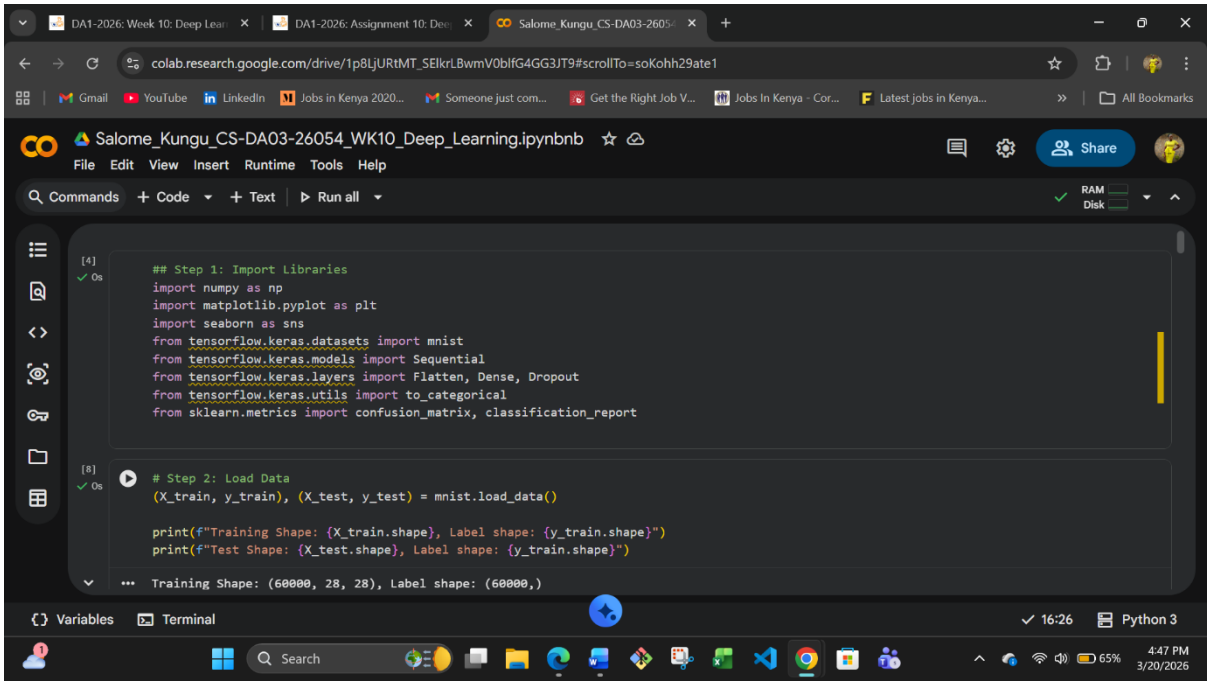
Task 1: Load and Explore the Dataset

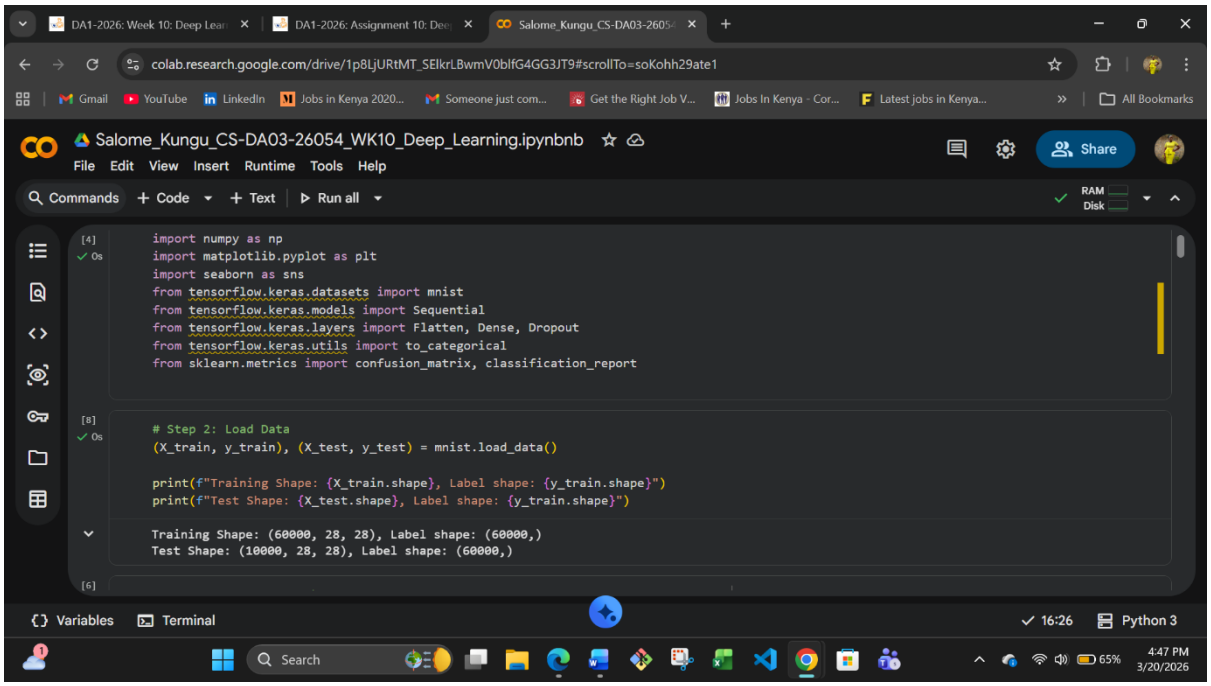
What:

- Loaded the MNIST dataset using tensorflow.keras.datasets
- Extracted training and testing data
- Visualized 9 sample images with labels

Why:

- To understand the dataset structure and image format
- To verify that images correspond correctly to their labels
- To gain intuition on handwritten digit patterns





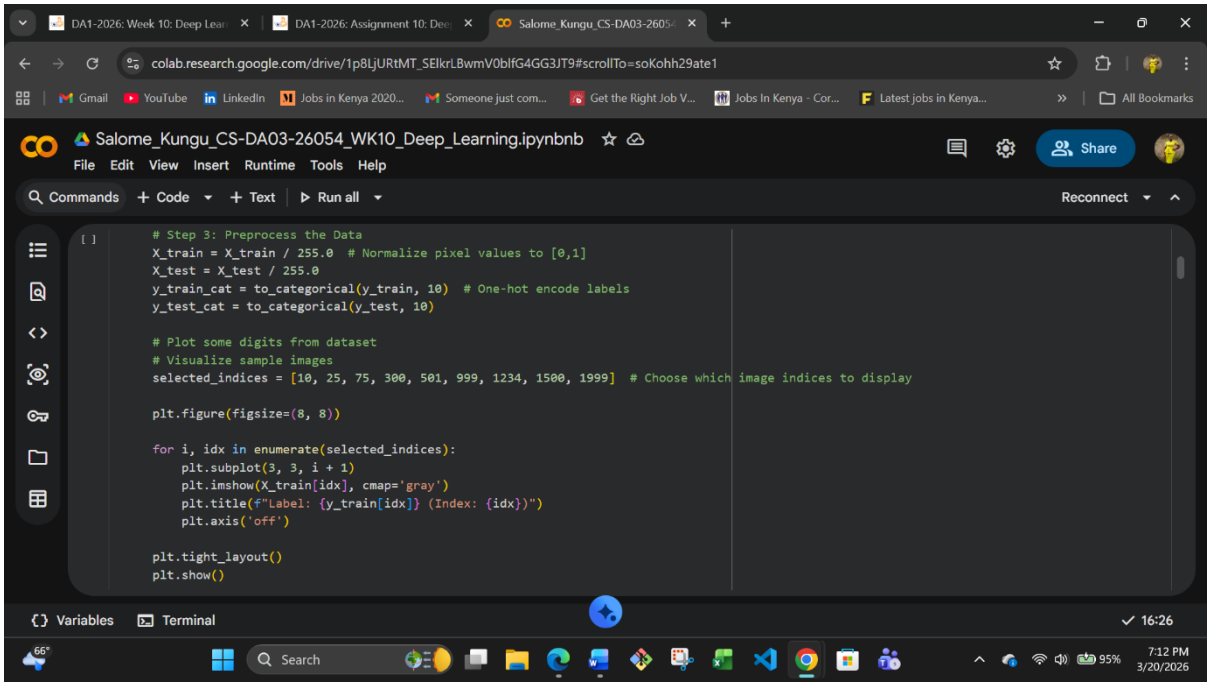
Task 2: Data Preparation and Preprocessing

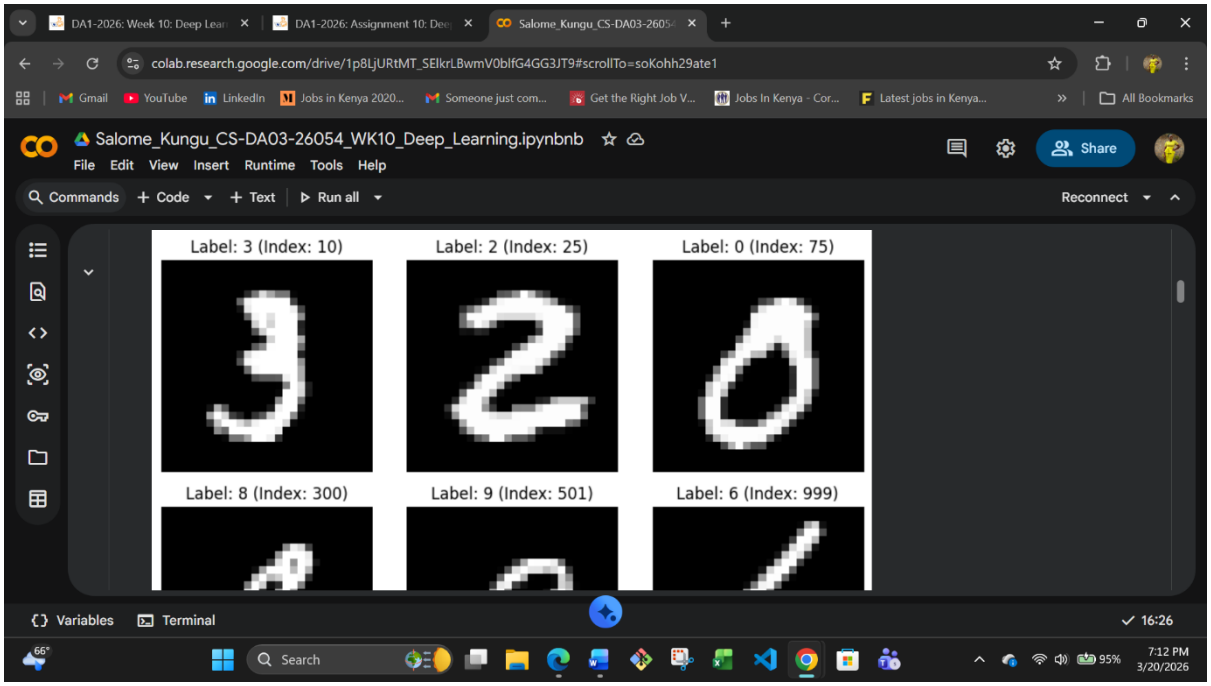
What:

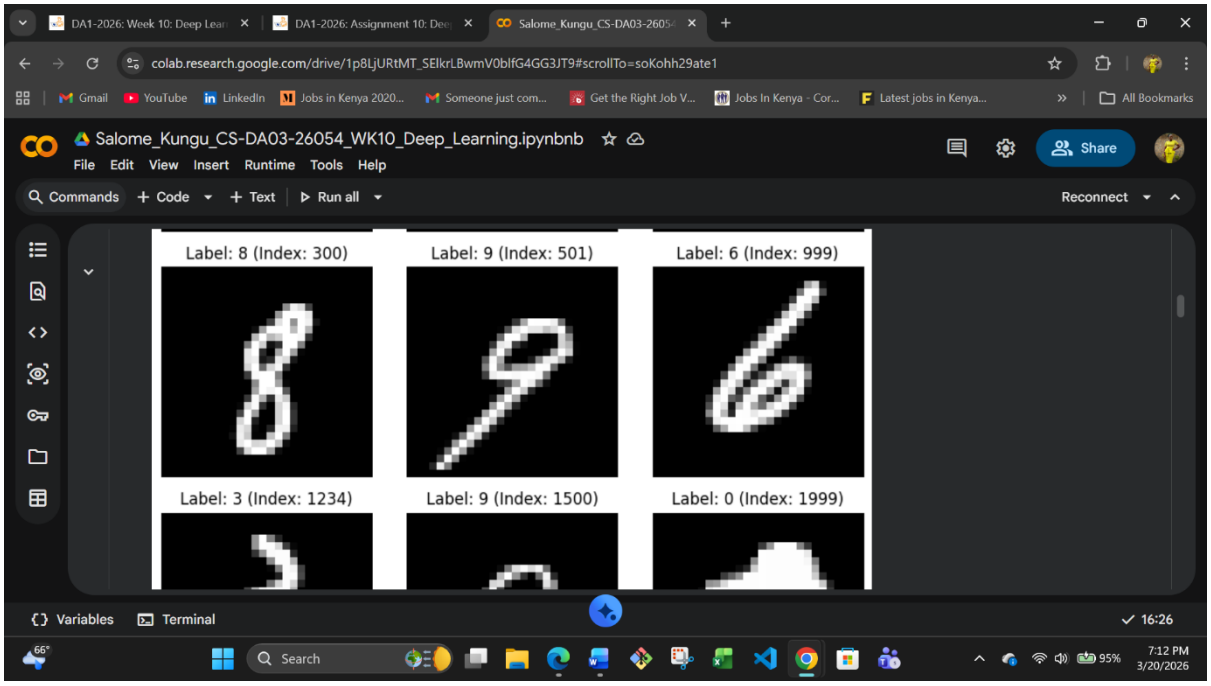
- Normalized pixel values to range [0,1]
- Applied one-hot encoding to labels
- Checked dataset shapes

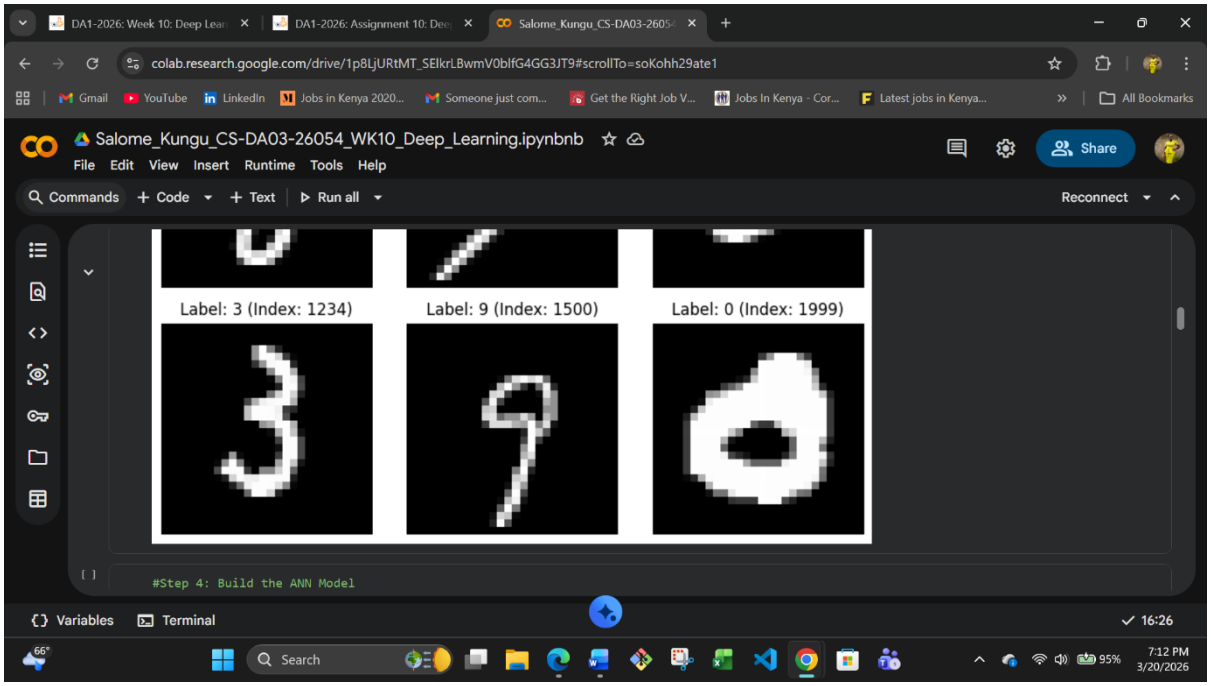
Why:

- Normalization improves convergence during training
- One-hot encoding is required for multi-class classification
- Ensures compatibility with neural network input









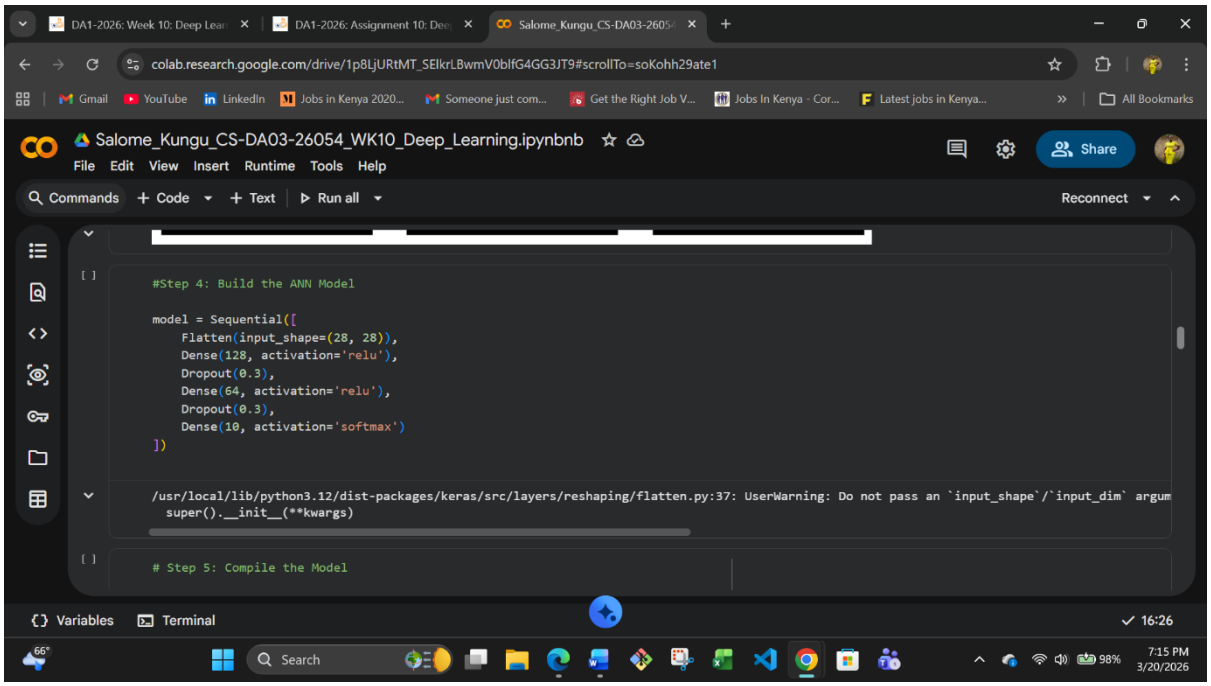
Task 3: ANN Model Construction

What:

- Built a Sequential ANN model with:
  - Flatten input layer
  - Two Dense hidden layers (128, 64 neurons)
  - Dropout layers (0.3)
  - Output layer (10 neurons, softmax)

Why:

- Flatten converts image into vector form
- Dense layers learn patterns in the data
- Dropout prevents overfitting
- Softmax outputs class probabilities



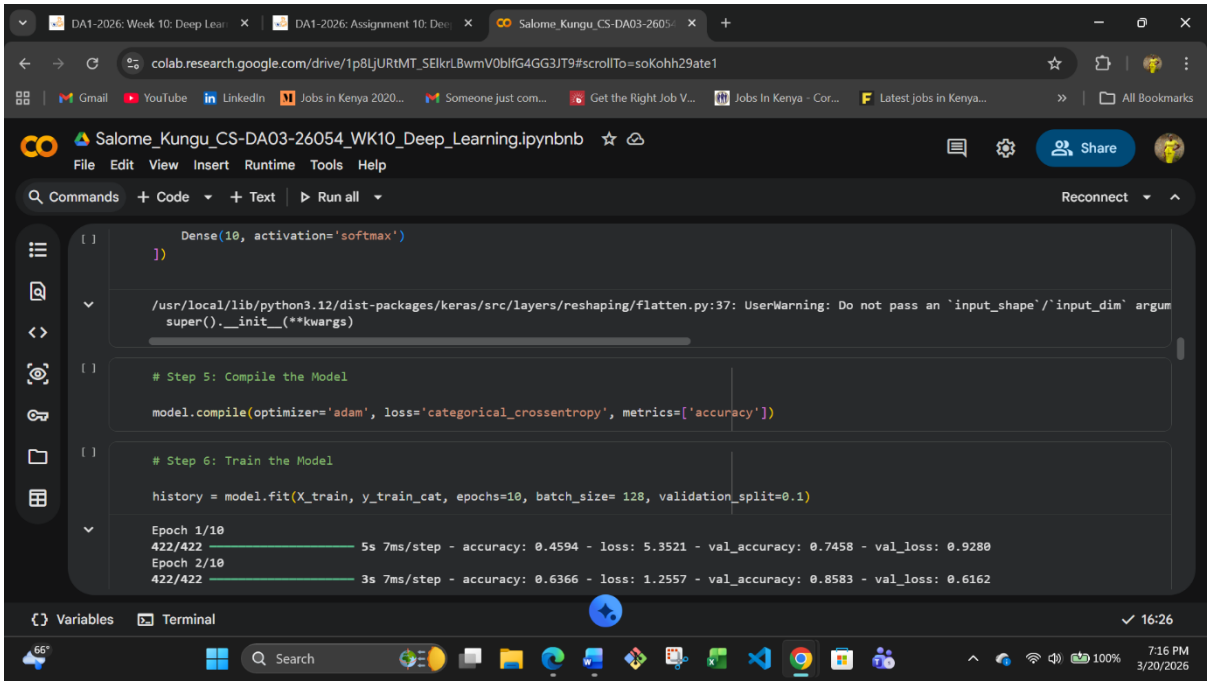
Task 4: Model Compilation

What:

- Compiled model using:
  - Optimizer: Adam
  - Loss: categorical_crossentropy
  - Metric: accuracy

Why:

- Adam provides efficient gradient optimization
- Crossentropy is ideal for classification problems
- Accuracy measures prediction performance



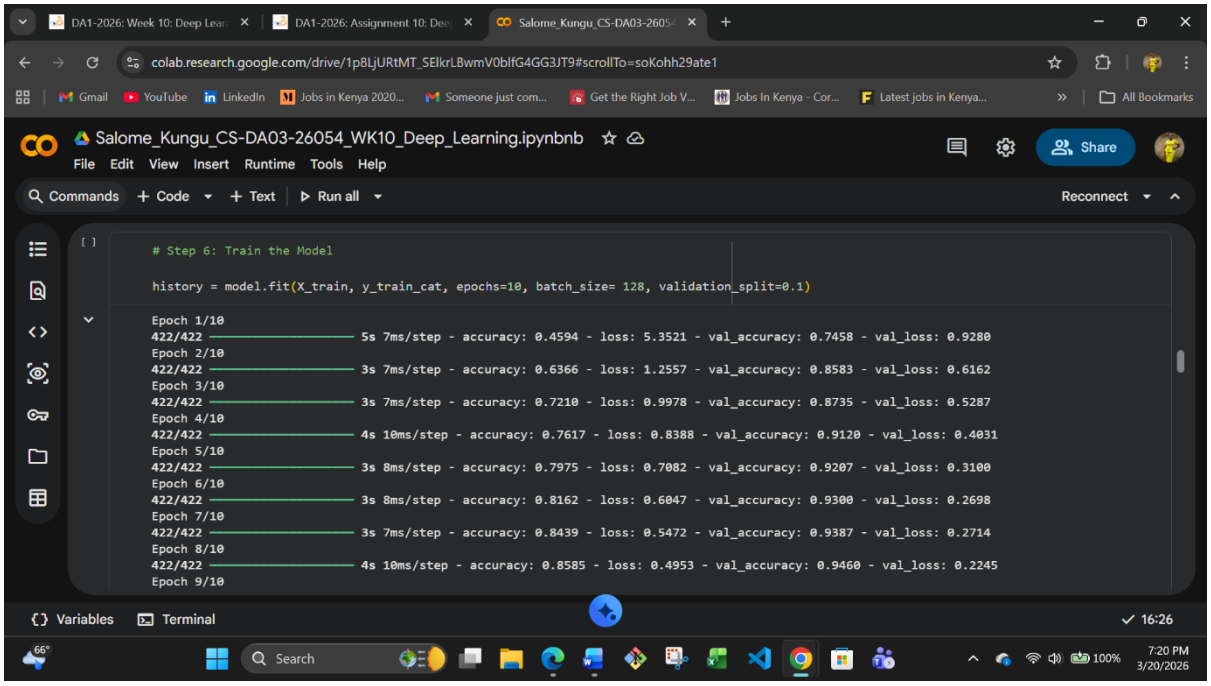
Task 5: Model Training

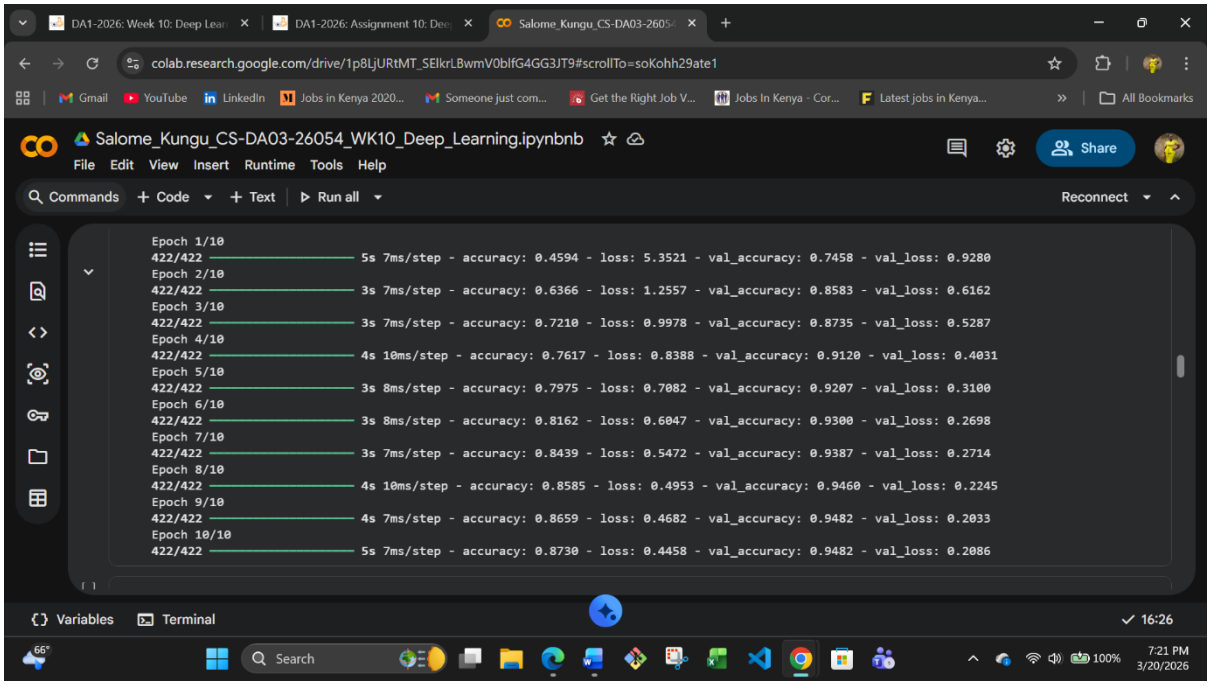
What:

- Trained model for 10 epochs
- Used batch size of 128
- Applied validation split of 0.1

Why:

- Training allows model to learn patterns
- Validation monitors generalization performance
- Batch processing improves efficiency





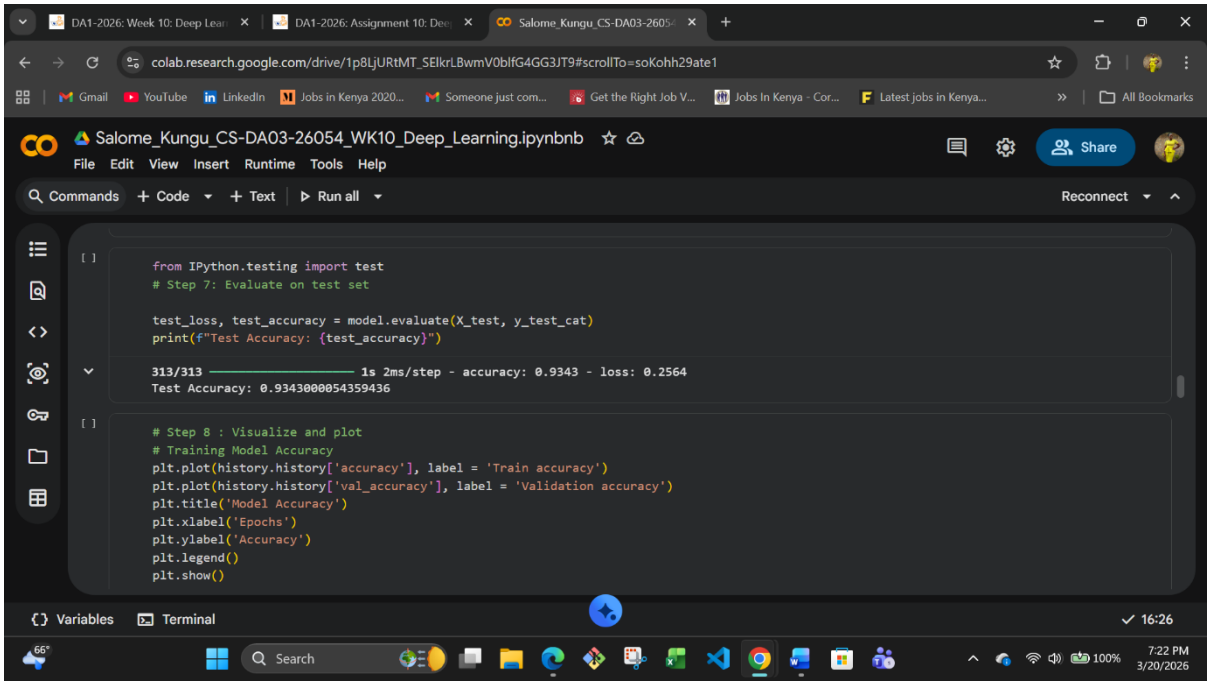
Task 6: Model Evaluation

What:

- Evaluated model on test dataset
- Reported final accuracy

Why:

- Measures model performance on unseen data
- Ensures model generalization



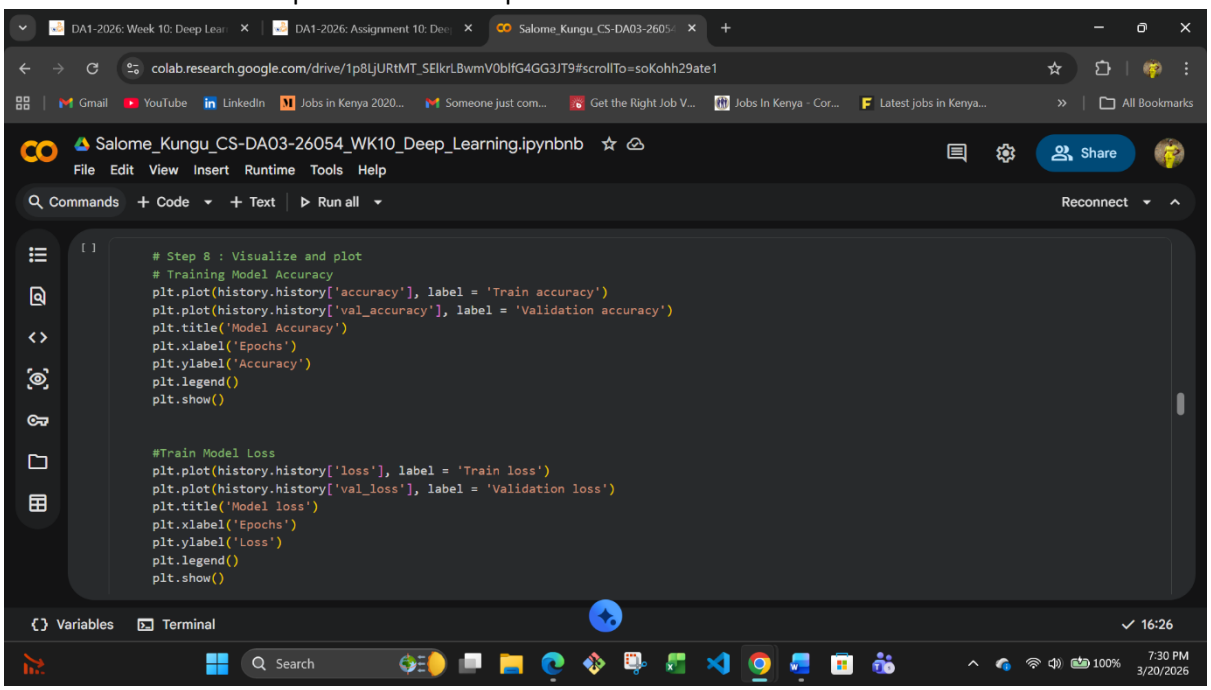
Task 7: Training Visualization

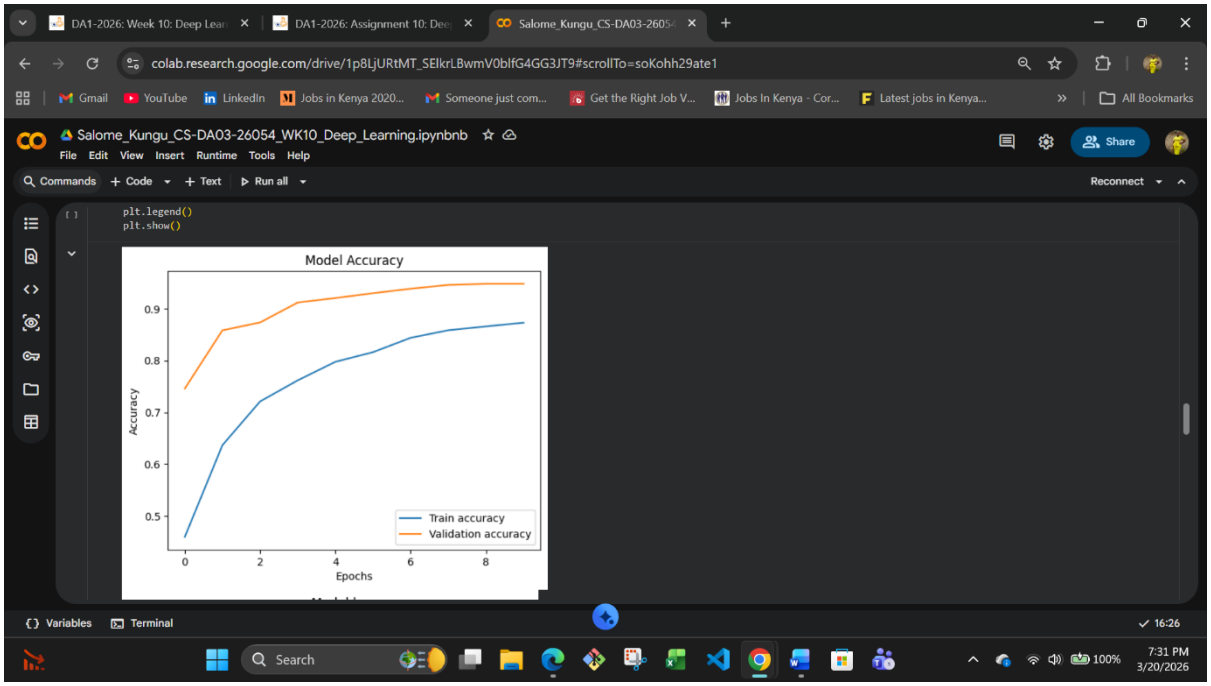
What:

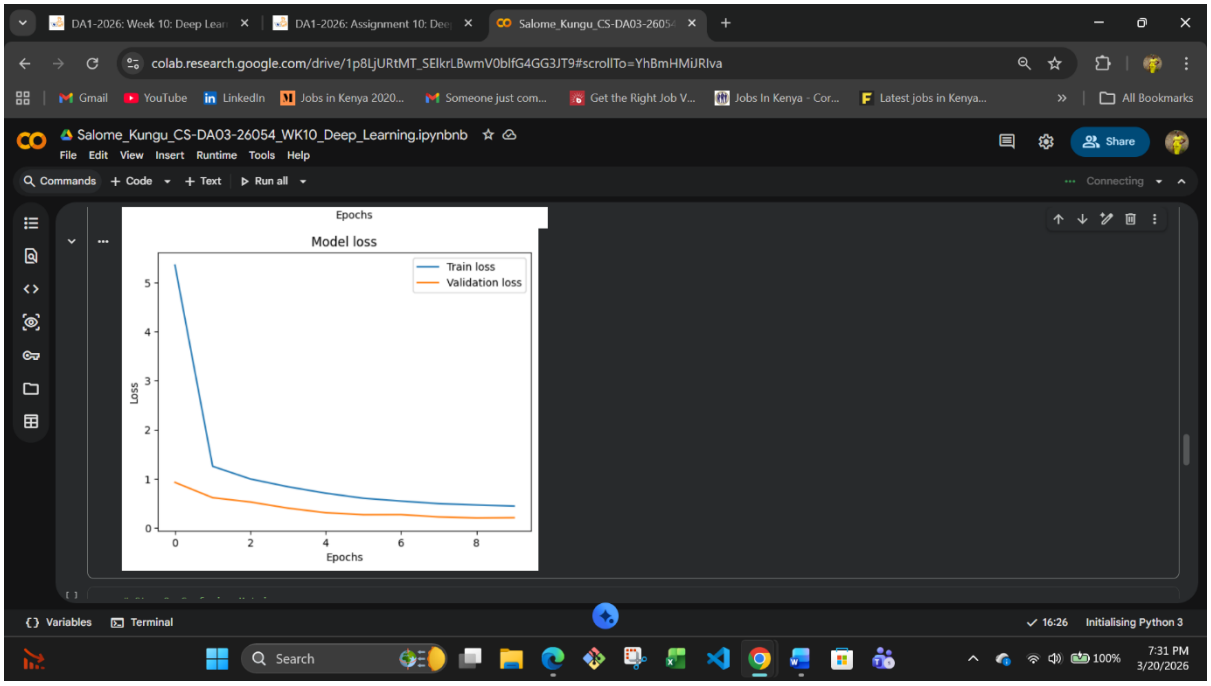
- Plotted training and validation accuracy

Why:

- Helps detect overfitting or underfitting
- Tracks model improvement over epochs







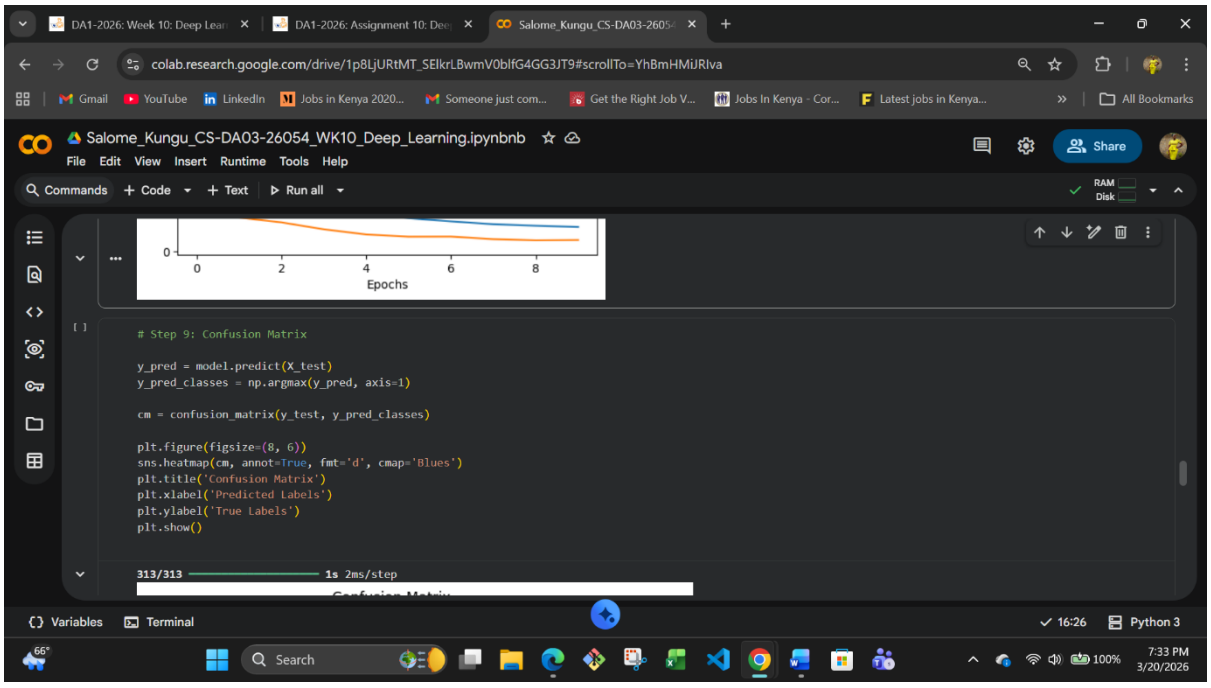
Task 8: Confusion Matrix

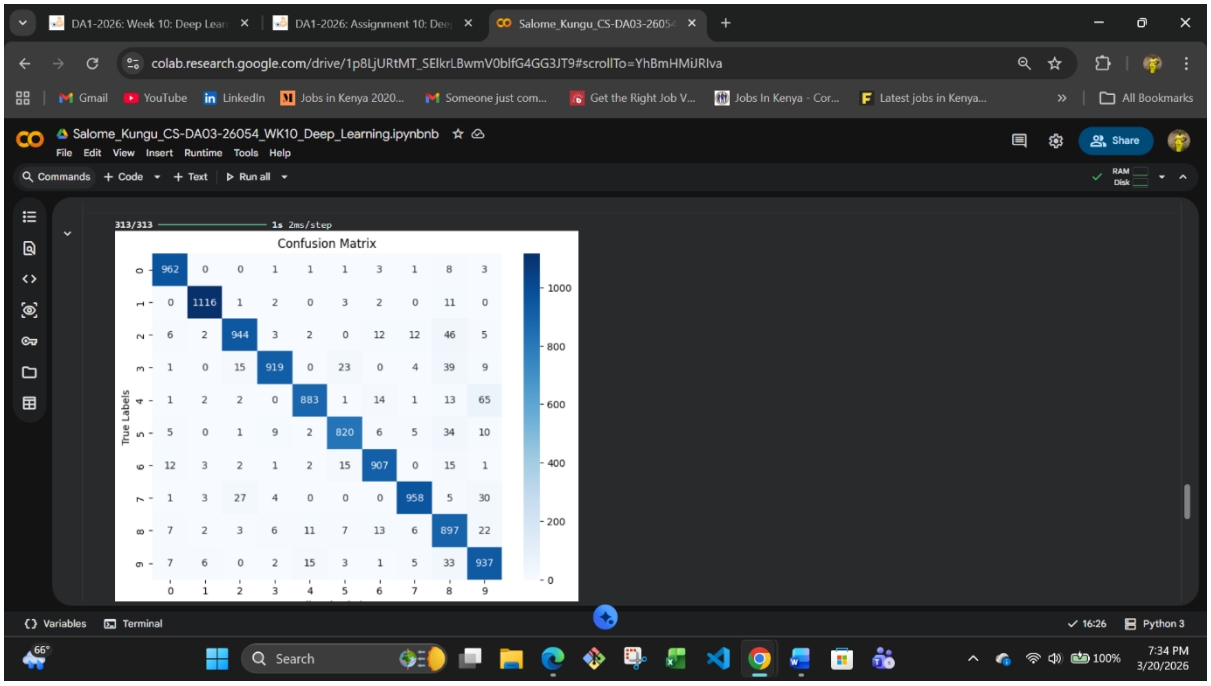
What:

- Generated confusion matrix using predictions

Why:

- Shows classification performance per class
- Helps identify misclassified digits





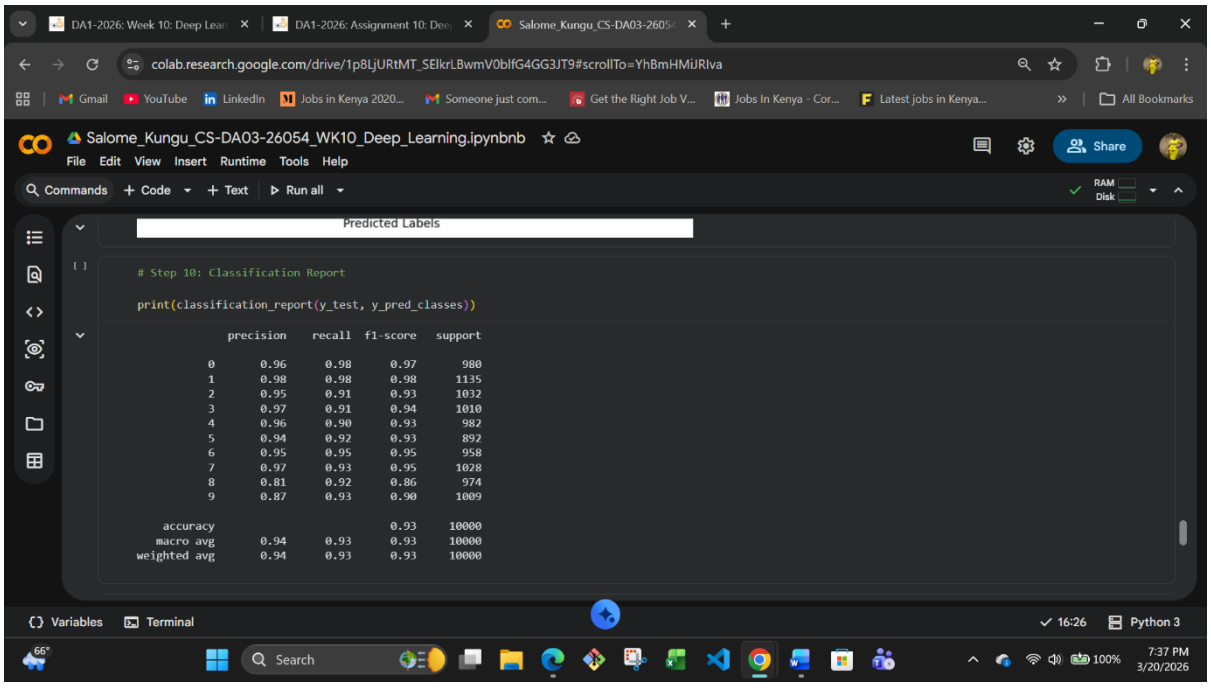
Task 9: Classification Report

What:

- Generated precision, recall, and F1-score

Why:

- Provides detailed performance metrics
- Helps evaluate class-wise accuracy



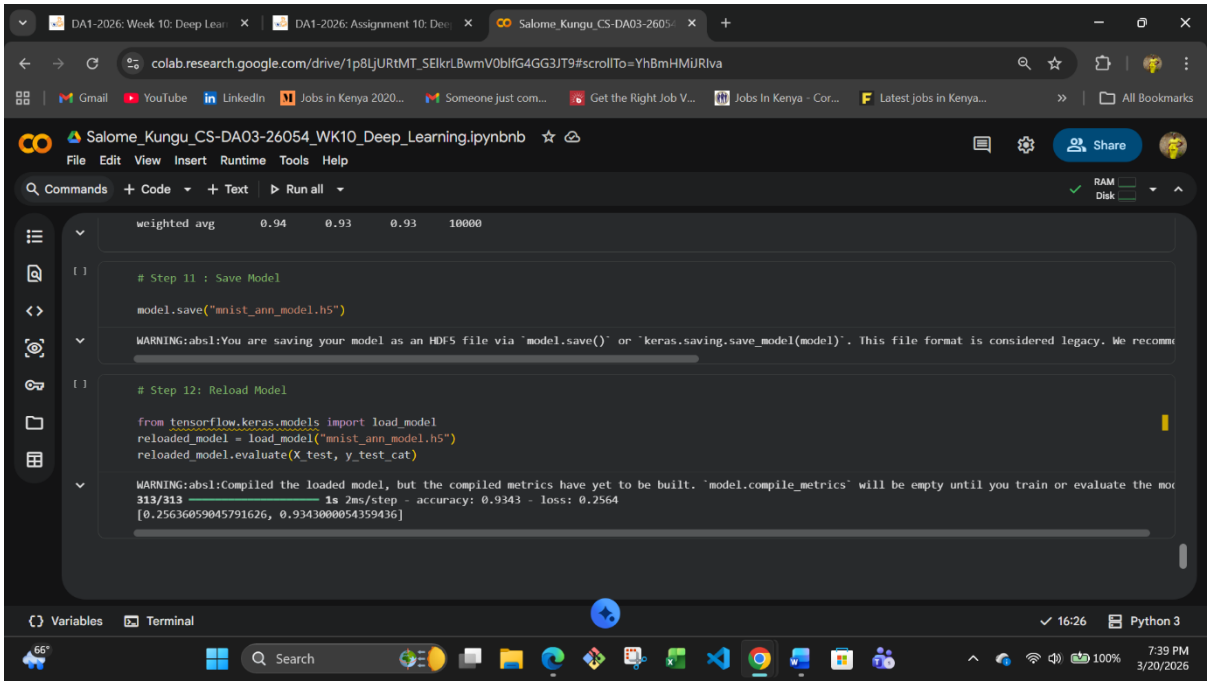
Task 10: Model Saving and Reloading

What:

- Saved model in Keras format
- Reloaded and evaluated model

Why:

- Enables reuse without retraining
- Supports deployment in applications



Conclusion

This assignment successfully demonstrated how to build a deep learning model for image classification using ANN. The model achieved high accuracy on the MNIST dataset and effectively classified handwritten digits.

Key skills developed include:

- Data preprocessing for images
- ANN model design using Keras
- Model evaluation and visualization